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AI Capital Pressure

Cross-industry and historical comparison of annual capital pressure in the AI industry chain

Stress testing under the EPR framework: comparison with normal industries and the same framework with the four historical infrastructure cycles dda | 2026-05-25 |_Non-investment advice | All data are updated to 2026 Q1 after the financial report

Data Verification and Version Notes

All key data in this article have been updated to the latest public sources available as of May 25, 2026. Major updates from earlier estimates:

Key Data	Updated Value	Source (Confidence)
Anthropic ARR	About $40B (May 2026 , further rising from April $30B)	Anthropic Official announcement $30B + TechCrunch Quoting insiders $40B (medium)
OpenAI ARR	$24-25B (April 2026)	Reuters/Sacra/OpenAI Q1 funding announcement (medium-high)
Microsoft AI ARR	$37B (FY26 Q3)	MSFT official conference call disclosure (high)
OpenAI Total commitments	About $1.4 trillion (5+ years cumulative nominal)	Sam Altman publicly confirmed (medium)
Big-5 hyperscaler 2026 capex	$670-750B (4 of 3 houses of Q1 raised later)	Q1 2026 10-Q + Company Guidance (High)
CoreWeave Quarterly Revenue	$2.08B (Q1 2026, +112% YoY)	CRWV 10-Q (High)
CoreWeave Revenue backlog	$99.4B (vs Early$30B+)	CRWV Q1 2026 Disclosure (High)
CoreWeave 2026 capex guide	$31-35B	CRWV guide (high)
NVDA Q1 FY27 Total Revenue	$81.6B (DC: $75.2B)	NVDA 8-K (High)
NVDA Q2 FY27 Guidelines	$91B (±2%)	NVDA Guidelines (High)
NVDA supply commitments	$95.2B (Q4 FY26 disclosure)	NVDA Q4 FY26 CFO commentary (high)

Key Data	Updated Value	Source (Confidence)
Anthropic Valuation/Financing	$380B (Series G Feb 2026) → $850-900B Expectations	Anthropic + TechCrunch Coverage (Medium-High)
OpenAI Valuation/Financing	$852B (March 2026 close $122B round)	Official Financing Announcement (High)
Goldman 2026 AI capex baseline	$765B	Goldman Sachs Research 2026-04 (High)

Of all the updates, the ones that have the biggest impact on the conclusion are Anthropic ARR jumping from $8-10B to $40B, Microsoft AI ARR disclosure $37B, and OpenAI commitments stack accumulating to $1.4T. This significantly changes the specific numbers of R_T terminal revenue and commitments/revenue ratio, but the overall conclusion direction of the EPR framework remains unchanged.

Central Conclusion

Conclusion first. This article uses the unified indicator EPR (Economic Pressure Ratio) = annual capital service pressure / verifiable terminal income to compare the AI industry chain with normal industries and historical bubbles under the same framework.

Let’ s first define it clearly: EPR is not a traditional “leverage ratio” (not Debt/Equity, not Debt/EBITDA), but a capital service pressure ratio—a measure of how much capital-related costs (depreciation + cash operations + interest) must be absorbed from terminal revenue each year. This metric is essentially a simplified form of Jorgenson's (1963) capital service cost theory and utility revenue requirement model, applied to AI cycle analysis. EPR is a nomenclature adopted for the convenience of discussion in this article; the framework itself is not new.

Under the assumptions of this article (K=$1.3T, weighted asset life 6.1 years, o=10%, r=5.5%, d=30%), the current EPR_cash of the AI industry chain is in the 1.6x–2.3x range, with a median of about 1.8–2.0x. These are stress test results based on explicit assumptions and are not precise industry measurements.

Put it into the benchmark and see:

- Normal industries (Utilities, Telecommunications, Mature SaaS, Automotive, Aviation) EPR approximately 0.65–0.95x. AI’ s current EPR is 2–3 times higher than normal industry benchmarks.
- Approximate estimation of historical infrastructure cycle peak EPR with the same framework: 1840s British Railways 2.5–3x, 1996–2001 U.S. telecommunications optical fiber (stripping out steady-state voice services, only looking at new IP/data segments) is about 3x, 1873 U.S. Railways is about 1.2x (aggregated caliber), 2014 U.S. shale oil is about 1.5–2x. 2007 US housing 0.9–1.1x, but housing is a price + credit quality + securitization structural bubble, and EPR is not a suitable indicator.

- AI's current EPR is orders of magnitude between 1873 US rail (1.2x) and 2014 US shale oil (1.5-2x), and lower than 1840s peak UK rail (2.5-3x) and 2000 new telecoms segments (approximately 3x). It is significantly lower than the median EPR I estimated in 2025 2.5-3.0x - this reflects the surge in Anthropic ARR in the first half of 2026 ($9B→$40B), Microsoft AI ARR ($13B→$37B), and the substantial expansion of R_T driven by the rapid growth of OpenAI.
- Warning required: EPR Center 1.8-2.0x remains well above normal industry benchmarks and remains in "infrastructure bubble magnitude" territory. AI orders of magnitude span the 1873 rail (1.2x) position but not yet the 1846 rail (2.5-3x) position.
- AI does not come close to 2007 US housing patterns. Housing is a "price + credit quality + securitization" structure, which is a different failure mode than AI's "capex front-loading + short asset life".

Capex to GDP (strictly distinguishing between level and growth contribution):

- AI capex level/GDP (stock caliber): Goldman estimates that AI capex currently accounts for about 0.8% of GDP. 2026 The total capex of Big-5 hyperscalers in the United States is $670-750B (of which ~75% is AI related = $500-545B), accounting for about 1.7-1.8% of U.S. GDP (~$30T) (including the industry-wide AI capex of about 2.0%). UK Rail capex to GDP in 1847 7-8%. U.S. telecom capex accounted for approximately 1.2% of GDP in 1999-2000. AI's current level is slightly above peak 2000 telecoms and well below 1840s railroads.
- AI capex contribution to GDP growth (incremental caliber, much more sensitive than horizontal caliber): St. Louis Fed estimates 2025 Q3 AI-related investment accounts for 39% of real GDP growth. Jason Furman estimates that investment in information processing equipment and software accounts for 92% of H1 2025 GDP growth. But Goldman Sachs (Jan Hatzius) countered: After deducting imported capital goods, the net contribution of AI capex to GDP growth is only 0.1–0.2 percentage points. The difference between the two calibers is more than 100 times. This article uses the St Louis Fed figures as a reference, but acknowledges that Goldman's imported capital goods adjustment is an important methodological correction.

Confidence. "AI EPR far exceeds normal industry benchmarks": High. "AI EPR Median 1.8-2.0x": Medium (strongly relies on R_T caliber assumptions; the greatest uncertainty in R_T estimates is that internal AI usage attribution and private placement ARR numbers are not audited). "AI orders of magnitude between 1873 railways and 1846 railways": Medium. "AI is likely to follow a digestion-like path rather than a complete collapse": Medium-High. "Specific clearing time point": low. All historical EPR numbers have low-medium confidence.

1. EPR Framework: What it is and what it is not

1.1 Definition

$P = K/L + o \times K + r \times d \times K$

$EPR_cash = P/R_T$

Among them, K = deployed capital stock, L = weighted average economic life, o = cash operation and maintenance / K, r = average debt financing interest rate, d = debt financing proportion, R_T = verifiable terminal revenue.

1.2 Framework traceability (not a new invention)

Strictly speaking, EPR is not a new metric. It is essentially a combination of several existing concepts:

- Economic level: Jorgenson (1963) Capital Service Cost Theory. Capital service price c = (i + δ - π) × p_K, where δ is the economic depreciation rate, i is the interest rate, and π is the asset price change. This is modern productivity accounting
- Basics of (BEA, BLS, KLEMS).
- Regulatory Economics: Utility revenue requirement formula. rate base × (depreciation rate + return on capital + O&M) = income that must be recovered. Standard practice in regulating ROE models for 60+ years.
- Real estate valuation: The reciprocal of cap rate (NOI / property value) is a similar indicator, 1930s already exists.
- Enterprise valuation: Equivalent to "implied required revenue to justify capex" analysis, which Damodaran and others have repeatedly used in the hyperscaler AI business.

The contribution of this article does not lie in the framework itself, but in applying it to the AI cycle + making a horizontal comparison with the current normal industry + with the four major infrastructure cycles in history. EPR is a convenient nomenclature adopted for the discussion in this article.

1.3 What is EPR

- EPR is capital service pressure ratio. Compress the capital-related costs (depreciation + cash operation and maintenance + interest) that the industry must absorb from terminal revenue every year into a single number.
- EPR is more information dense than Debt/Revenue or Capex/Revenue because it also incorporates existing capital pressure, operating pressure and financing costs.
- EPR can be used for cross-industry and cross-period comparisons within the same framework (although differences in historical accounting standards are inevitable).

1.4 What EPR is not

- EPR is not a traditional leverage ratio. It is not Debt/Equity, Debt/EBITDA, Assets/Equity, Net Debt/Revenue. Calling EPR "leverage" will make readers mistakenly think that it measures debt risk, but many of the core risks of AI are not debt, but equity dilution, depreciation eating up profits,

cloud contract renegotiation, capex deceleration leading to valuation compression, and GPU residual value revisions.

- EPR is not a complete profit margin indicator. In this article, P only includes depreciation + opex + interest at the infrastructure layer, and does not include R&D, SG&A, sales, customer service, equity incentives, data procurement, promotion subsidies, and taxes at the model company and application layer. Real industry profit pressure is higher than the EPR suggests.
- EPR > 1 is not directly equivalent to "the industry is not profitable". But the terminal revenue is not enough to cover the annual infrastructure pressure defined in this article.
- EPR_cash defaults to cash interest ($r \times d \times K$), without charge equity capital opportunity cost. If WACC is used instead, the EPR increases significantly (see sensitivity analysis).
- Historical EPR is an approximate estimate of the same frame, not a precise measurement of "the same caliber". All historical EPR numbers have low-medium confidence.

Interpretation interval of 1.5 EPR

EPR_cash area room	Interpretation	Typical examples
< 0.6x	Low capital pressure, steady industry profitability	Consumer goods, Internet platforms, light asset services
0.6–1.0x	Medium capital pressure, most normal industries	Utilities, telecommunications, SaaS, automobiles, aviation
1.0–2.0x	High capital pressure, demand extrapolation or cycle high	Peak shale oil period, AI current (updated), 1873U.S. U.S. rail aggregation
2.0–4.0x	Infrastructure bubble magnitude, capitalize future needs in advance	1840s railway, 2000 new telecom optical fiber section
> 4.0x	Extreme situations, usually appearing in the last stage before a crash	A few extreme cases

2. Current EPR (data update) of the AI industry chain

2.1 Input parameters (update)

Parameter	Value (update)	Change from earlier estimate
K (deployed + AI capex stock under construction)	$1.3 trillion	Upgraded 8% from $1.2T (capex acceleration)
L (asset-weighted average economic life)	6.1 years	unchanged
o (Cash opex/K)	10%	No change

Parameter	Value (update)	Change from earlier estimate
r (average debt financing interest rate)	5.5%	unchanged
d (proportion of debt financing)	30%	unchanged
R_T wide diameter	$225B	Significantly increased from $150B (+50%)
R_T strict caliber	$160B	Significantly increased from $90B (+78%)
m_T (Terminal AI gross profit margin)	50%	Unchanged

Reason for K increase: 2025 hyperscaler AI capex about $260B + 2026 H1 deployment ~$260B + Neocloud + other cumulative deployments of subjects + construction in progress.

Core reasons for the significant R_T upgrade: Anthropic ARR rose from $9B at the end of 2025 to ~$40B at 2026 May (increased 4.4x); Microsoft AI ARR disclosed $37B; OpenAI maintained ~$25B but continued to grow rapidly. The rapid realization of AI terminal revenue is the most important variable for EPR improvement.

2.2 EPR Calculation (Updated)

Project	Formula	Results (updated)
Annual depreciation (stratified sum)	$\Sigma K_i/L_i$ = $1.3T × 16.4%	$213B
Annual cash opex	o × K	$130B
Annual cash interest	r × d × K	$21.5B
Annualized Economic Pressure P_cash	Subtotal	$364B
EPR_cash wide diameter	P/R_T(wide)	1.62x
EPR_cash strict caliber	P/R_T(strict)	2.28x
EPR_cash median estimate		about 1.8–2.0x
R_req	P/m_T	$728B
Current gap (width)	R_req − R_T	About $503B
Current gap (strict)	R_req − R_T	About $568B

Interpretation. EPR is 1.62x in wide aperture and 2.28x in strict aperture. Median estimate 1.8-2.0x. This is a significant downward shift from the mid-year 2025 estimate of 2.5-3.0x.

What has changed? R_T expanded significantly. This is the entire driver of EPR improvement. From mid-year 2025 to May 2026, AI terminal revenue roughly doubled, while P (capital pressure) only increased by about 8% (K rose from $1.2T to $1.3T). R_T growth rate >> P growth rate → EPR decreases.

What hasn't changed? All buffering and amplification factors are still present. OpenAI's total commitments have increased to $1.4T, CoreWeave backlog has increased to $99.4B, NVDA's recurring

investment scale >$40B, and hyperscaler 2026 capex guidance is still being revised upward. The AI cycle is accelerating in two directions at the same time: revenue is chasing, and capex is also chasing. EPR is the difference between the two races.

2.3 R_T bottom-up decomposition (core update table)

R_T is the most sensitive input of the EPR framework. The table below gives the item-by-item breakdown of 2026 May's strict and wide apertures.

Category	Strict Caliber	Wide Caliber	Confidence/Explanation
Anthropic (Claude API + Subscription + Claude Code)	$40B	$40B	TechCrunch cited insiders May 2026 (medium); Anthropic public $30B April 2026 (high)
OpenAI (API + ChatGPT Subscription)	$25B	$28B	Reuters reports / Sacra (Medium-High)
Microsoft AI Net($37B less	$27B	$30B	MSFT FY26 Q3 Disclosure

Category	Strict Caliber	Wide Caliber	Confidence/Explanation
~$10B Azure-OpenAI resale)			$37B ; This article deducts Azure-OpenAI resale points
xAI, Mistral, Cohere, other model companies	$5B	$8B	Low-Medium
Google Gemini Advanced + Workspace AI	$5B	$8B	Medium-Low (not separately disclosed by GOOG)
Dedicated AI SaaS (Cursor, Glean, Harve y, Perplexity, Cognition, Sie rra)	$15B	$20B	中
AI add-on incremental for traditional SaaS (Salesforce Agentforce, ServiceNow Now Assist)	$10B	$15B	Low-Medium (incremental only)
China AI commercialization (Baidu, Byte, Alibaba, Zhipu, DeepSeek, Moonshot)	$10B	$15B	Low
AWS Bedrock + GCP Vertex AI Terminal Billing (deduct Anthropic/OpenAI resale)	$5B	$10B	Medium-Low
	0 (not counted)	$35B	

Category	Strict Caliber	Wide Caliber	Confidence/Explanation
Hyperscaler Internal AI usage attribution (search, advertising, recommendation increment)			Low (not independently verifiable, but the scale is real)
Consulting/Professional Services/ In-house productivity savings	0 (not counted)	$10B	Low
Long tail (other paid)	$5B	$8B	Low
Exclusion 1: Azure/AWS/GCP sales Intermediate revenue to OpenAI/Anthropic	Exclusions	Exclusions	Avoid double counting
Exclusion 2: NVIDIA GPU Sales Revenue	Exclusion	Exclusion	Upstream Capital Goods Revenue
Total (rounded)	About $147B → Take $160B	About $227B → Take $225B	

The difference between the two calibers. Strictly speaking, only verifiable pure AI product revenue (API, subscription, Claude Code terminal products) will be included. Broadcast joins hyperscaler internal AI usage attribution (search, advertising, recommendation AI delta) and enterprise productivity savings. The main difference between the two is whether the attribution can be trusted.

Source of Anthropic $40B. The company officially disclosed in April that ARR had exceeded $30B (early April). TechCrunch follow-up reported ARR "closer to $40B" per internal source in late April. This figure has not been audited and has not been disclosed in SEC filings. This article gives a medium confidence level. The trajectory from $14B (Feb) → $19B (March) → $30B (April) → $40B (May) is consistent with the 6-week doubling rate.

Microsoft $37B net $27B split logic. Microsoft Official FY26 Q3 Disclosed AI Business ARR $37B (+123% YoY). But this number includes the portion of Azure OpenAI Service sold to the end customer (already billed at the end customer and can be included in R_T) and the resale portion sold to OpenAI itself (which should be excluded to avoid duplication with OpenAI ARR). This article estimates the overlap to be approximately $10B, netting $27B. This split estimate is subject to uncertainty of ±$5B.

Three, K, annual Capex and Commitments are strictly stratified

The AI industry chain involves three types of related but essentially different numbers. Mixing up leads to serious double counting or conceptual errors.

Level	Definition	Accounting Nature	Use in this article	Typical2026Number (Updated)
K : Deployed + Assets Under Construction Capital Stock	Assets on Book NBV + Construction in Progress Stock	Enter EPR Numerator	Estimate $1.3T	
Annual Capex	The current year’ s PP&E acquisition flow	Flow	Forecast future K ; used for capex/GDP level	Big-5 $670-750B; with AI ~$520B
Commitments	Contractual obligations not yet fully deployed Obligations/announced	Contractual obligations Obligations/ announce d	Future K leading indicators	OpenAI Total stack~$1.4 Trillion over 5+ years

3.1 K (about $1.3T, updated with disassembly)

- Hyperscaler AI cumulative capex stock (Big-5 2022-2026 cumulative by AI portion 75%): approx. $780880B
- Neocloud AI capex stock (CoreWeave PP&E $36.4B + Lambda + Crusoe + Nebius, etc.): approx. $90-130B
- Others (Oracle AI infra, self-built by enterprises, government/sovereign investment): $200-300B
- The total median is about $1.3T

3.2 Annual Capex (2026 latest guidance)

Subject	2024	2025	2026Guidelines/ Estimates (Updated after Q1)	AI Part (~75%)
Microsoft	~$56B	~$95B	~$190B (including $25B high component pricing)	~$140B
Alphabet	~$53B	~$75B	$180–190B (raised from $175-185B after Q1)	~$140B

Subject	2024	2025	2026Guidelines/ Estimates (Updated after Q1)	AI Part (~75%)
Amazon	~$78B	~$105B	~$200B	~$150B
Meta	~$37B	~$65B	$125–145B (raised from $115-135B after Q1)	~$100B
Oracle	~$7B	~$25B	~$60B	~$50B
Big-5 Total	~$231B	~$365B	$755–805B (CreditSights	~$580B

Subject	2024	2025	2026Guidelines/ Estimates (Updated after Q1)	AI Part (~75%)
			$750B)	
Neocloud total	~$15B	~$35B	~$60-70B	~$65B
Industry AI capex Traffic		~$400B	~$645B	

Sources: SEC 10-Q Public Disclosures, Q1 2026 Earnings Call Guidance, CreditSights/Goldman Sachs/ Yahoo Finance Comprehensive (updated after Q1). Four of the Big-5 companies raised their capex guidance after Q1; only Amazon maintained its original plan (because it was already the highest in the industry).

Observation: Goldman Sachs latest baseline model estimates 2026 AI capex $765B, consistent with CreditSights $750B estimates. Goldman predicts AI capex of $1.6T in 2031. Capital expenditures/ revenue ratios have exceeded Oracle 86%, Meta 54%, Microsoft 47%, Alphabet 46%, Amazon 25% - historically unimaginable levels.

3.3 OpenAI Commitments stack (updated to $1.4T)

Sam Altman publicly confirmed that OpenAI’ s total commitments are approximately $1.4 trillion (covering the next 5+ years). This is the largest single point in the industrial chain. The table below is the breakdown.

Contract/Commitment	Nominal Size	Time Window	Confidence
Oracle compute deal	About $300B	5 years (2027-2031)	High (Oracle CEO Safra Catz confirmed many times)
Stargate (OpenAI/ SoftBank/Oracle/ MGX)	About $500B (announced $400B+ committed)	4 years (2025-2028)	Medium (announced , overlapping with Oracle part)

Contract/Commitment	Nominal Size	Time Window	Confidence
AMD (6 GW Instinct GPU)	Approximately $90B Cumulative Hardware Revenue	From H2 2026 Onwards	Medium (AMD/OpenAI Public Disclosure)
NVIDIA investment	up to $100B	Conditional, gradually deployed per GW	Medium (announced strategic cooperation)
Amazon AWS	$38B	7 years (2025-2031)	High (2025-11 signed by both parties)
Microsoft Azure (after reorganization)	about $250B cloud purchase	many years	medium (reorganization agreement between both parties)
CoreWeave	Approximately $22.4B	to 2032	High (CRWV 10-Q Disclosure)
Broadcom Self-developed chips + Others	About $50-100B	Many years	Mid
Total Notional Stack (Altman Public Numbers)	Approximately $1.4 Trillion	5+ Years	Mid-High
OpenAI 2026 May	~$25B	current	medium-high

Contract/Commitment	Nominal Size	Time Window	Confidence
ARR			
Commitments / Revenue Ratio			About 56x

The OpenAI commitments/revenue ratio rose to 56x from an early estimate of 28-36x. This is a level that has not been achieved by any model company or customer/supplier relationship in history. Even if OpenAI ARR grows at 100% CAGR (already extremely optimistic), it will take 5 years for the commitments/revenue ratio to drop below 5x.

Compare Anthropic commitments stack:

Contract/Commitment	Nominal Size	Description
Amazon	$5B Cash + Up to $25B + 5GW Trainium + $100B AWS Procurement (10 years)	Primary Training Partner
Google	$10B Cash + Up to $40B + 5GW TPU	Announced April 24th
NVIDIA	Up to $10B + 1GW GPU supply	
Microsoft		

Contract/Commitment	Nominal Size	Description
	Up to $5B + Anthropic Buy $30B Azure compute	
Broadcom	$21B AI Revenue (Anthropic 2026 estimate)	Mizuho estimates
U.S. compute infrastructure Investment Commitment	$50B	Nov 2025Public
Total nominal stack estimate	$200-300B+	Relative ARR $40B vs. 5-7.5x

Anthropic's commitments/revenue ratio (5-7.5x) is much lower than OpenAI (56x). This reflects two different business models: Anthropic’ s revenue is expanding rapidly (doubling in 6 weeks), and its commitment scale is relatively restrained; OpenAI’ s commitment scale is extremely large, and it relies on future revenue to deliver.

4. Normal industry EPR benchmark (rough estimate, non-audit industry model)

The normal industry EPR range serves as the benchmark. All figures are based on industry average estimates based on 2024-2025 public financial reports, with a medium confidence level. Warning: Industry aggregate data is rough. Company income, D&A, opex, and interest come from different companies and accounting standards, and it is impossible to have the same caliber. These numbers are order of magnitude benchmarks, not exact comparisons.

Industry	Estimated EPR_cash	Asset life characteristics	Risk characteristics
Mature SaaS	0.69x	Short (3–5y)	Light asset, opex adjustable
Traditional telecommunications (post-2002)	0.76x	Medium long (10–15y)	Steady-state industries
Utilities (regulated)	0.80x	Long (30+y)	High debt + stable returns
Aviation	0.90x	Medium (15–25y)	Close to the equilibrium line
Automotive OEM	0.92x	Short to medium (5–10y products)	Close to balance line

Industry	Estimated EPR_cash	Asset life characteristics	Risk characteristics
AI (updated wide aperture)	1.62x	Hybrid (4–15y)	Far beyond normal baseline
AI (strict caliber after update)	2.28x	Same as above	Far beyond the normal baseline

The normal industry median EPR is about 0.80x, and even high capital intensity industries (utilities, automobiles) rarely exceed 1.0x. AI's current EPR is 2-3 times that of normal industries. Still an order of magnitude away from normal benchmarks.

5. Approximate estimation of historical bubbles in the same framework

Use the same formula to construct approximate EPR estimates for four historical bubbles. All figures are based on published academic research and economic history, accounting and - statistical caliber are different from modern times, confidence level is low - medium. Only for order of magnitude positioning.

5.1 1844-1847 British Railmania

background. Rail capex accounted for 7-8% of UK GDP in 1847 (Mitchell historical statistics) and twice as much as military spending. 1844-1846 Railroad stock index doubled before crashing, 1847-49 shares fell 50%+.

Item	Estimate	Description
Cumulative rail capital commitments1844-1847	£170M	Odlyzko estimate
UK GDP 1846	£550-600M	Mitchell historical statistics
1847 Railway capex/GDP	7-8%	Tandfonline (2022) Historical Research
Years D&A (L=30y on £170MDepartment Department)	£5.7M	Long life of railway assets
year opex(K 10-15%)	£17-25M	
Annual interest (d=60%, r=4.5-5%)	£4.6-5.1M	
P_cash	£27-36M	
1846Railway operating income	£10-15M	
Estimated EPR_cash	2.0-3.5x	Median is about 2.5-3x

Interpretation. 1840s UK Rail EPR estimate 2.5-3x. The mechanism is very similar to AI: real technology + real future demand + capital market extrapolation in advance + reflexive financing. Result: 1847 Bank of England Interest rate hikes and banking crisis trigger collapse. Railroad stocks fell 1847-1850 50-80%. But the rail network was eventually built and produced ongoing returns—investors lost money and society gained.

5.2 1865-1873 US Railroad Expansion

background. 1865-1873 U.S. rail mileage doubled from 35,000 to 70,000 miles. In September 1873, the bankruptcy of Jay Cooke triggered the Panic of 1873. 1873-1879 Long Depression, 89 railroads (about 25%) went bankrupt.

Item	Estimate	Description
Accumulated railroad capital1865-1873	$2.7B	historical estimates
US GDP 1873	approximately $7B	Williamson historical

Item	Estimate	Description
Cumulative capex/GDP	About 38%	Extremely high
Year D&A (L=30y)	$90M	
year opex(K 15%)	$405M	
Annual interest (d=70%, r=7%)	$132M	High leverage and high interest rate
P_cash	$627M	
Annual Railway Operating Revenue1872	Approximately$520M	Census/ICC historical
Estimated EPR_cash (aggregated)	Approximately 1.2x	Less than UK figures

Interpretation. Overall rail EPR approximately 1.2x. Profits on old mature lines masked extreme pressure on new speculative lines. Crash Mechanism: Jay Cooke Fundraising Failure → Credit Crunch → Overall Railroad Bond Price Collapse → Bank Failures → 6 Years of Long Depression.

5.3 1996-2001 US Telecom Fiber Optic Foam

background. WorldCom CEO claims that Internet traffic doubles every 100 days (actually about 2x per year). In 1996-2001, U.S. telecommunications companies issued $500B+ of new debt and laid 80M miles of optical fiber. Global Crossing 2002 Bankruptcy leaves $12.4B debt. WorldCom 2002 Bankruptcy, largest U.S. accounting fraud. 2002 Only 2.7% US fiber is "lit". 2004 Long-distance wholesale prices fell 55% year-on-year.

Item	Estimate	Description
Accumulated US telecom capex 1996-2001	About $500B	Industry aggregation
Peak year capex (2000)	$120B	BEA
US GDP 2000	$10.25T	BEA
Peak capex/GDP	About 1.2%	
Year D&A (L=15y on $500BDepartment Department)	$33B	
year opex(K 10-12%)	$50-60B	
Annual interest (d=60%, r=8%)	$24B	
P_cash	$107-117B	
2001 IP/data terminal revenue	$30-50B	Split from total telecom revenue
	About 2.5-4x	Median 3x

Item	Estimate	Description
Estimated EPR_cash (IP/data new segment only)		
Industry-wide (including voice steady-state) EPR	About 0.5-0.8x	Steady-state business covers the pressure of new IP

Interpretation. After stripping off the steady-state voice service, the EPR of the new IP/data segment is about 3x. This is almost completely parallel to the current situation of AI: Hyperscaler's overall business (search, advertising, e-commerce, Office) is healthy, but the newly built AI infrastructure is under great pressure alone.

5.4 2005-2008 American Housing (Contrast Case)

Item	Estimate	Description
Home Mortgage Debt/GDP 2007	97%	FRED
Total value of housing stock2007Q3	Approximately$24T	
Home Mortgage Debt2007Q3	Approx.$10T	FRED HHMSDODNS
Year D&A (Housing2%Depreciation)	$480B	
Annual cash opex	$700B	3% of about K
Annual interest	$600B	
P_cash	about$1.78T	
Annual rental income (actual + imputed)	Approx. $1.7-2T	BEA
Estimated housing EPR_cash	0.9-1.1x	

Interpretation. Housing EPR is about 1.0x, not extreme. The core of the housing bubble is not in EPR, but in three different dimensions: (1) price/income ratio rising from 3x to 5x; (2) credit quality (NINJA loans, option ARM); (3) securitization deregulation and rating fraud. Important conclusions. EPR is not a be-all and end-all indicator for all bubbles. The housing bubble is a three-layer structure of "price + credit quality + securitization", and the railway/telecom/AI is a "capital front + demand extrapolation + short asset life" structure.

5.5 Other historical cases (short list)

Case	Period	EPR estimate	Main rupture mechanism
Japanese asset bubble	1985-1990	n/a (price bubble)	The stock market fell 60% , land prices fell 80%, 30 years of deflation
China’ s steel surplus	2010-2016	About 0.5-0.7x	Price falls70%, supply-side reform and integration

Case	Period	EPR estimate	Main rupture mechanism
U.S. shale oil	2011-2016	About 1.5-2x	Oil prices collapse 75%, ~100 E&Ps bankrupt
Cryptocurrency (2017/2021)	2017, 2021	n/a (narrative bubble)	Bitcoin fell 80% multiple times

6. Main comparison table: AI’ s position in the leverage spectrum (updated)

Put all previous calculations in a table.

Case	Period	EPR Approximate Estimate	Asset Life	Terminal Crash Characteristics	Similar to AI Degree
Mature SaaS	Current	0.69x	3-5y	n/a (steady state)	Low
Traditional Telecommunicatio ns (post-bubble)	Current	0.76x	10-15y	n/a	Low
Utilities	Current	0.80x	30+y	n/a	Low

Case	Period	EPR Approximate Estimate	Asset Life	Terminal Crash Characteristics	Similar to AI Degree
regulated					
Aviation	Current	0.90x	15-25y	Cyclic Bankruptcy	Low
Auto OEM	Current	0.92x	5-10y	Cyclic Bankruptcy	Low
China Steel	2015 Peak Value	0.5-0.7x	20+y	Price fell70%	Low
US Housing	2007Peak Value	0.9-1.1x	50+y	House prices fall35%, financial crisis	Low (structure different)
U.S. Railroads	1873 Peak Value	Approximately 1.2x (aggregated)	50+y	25% Industry Bankruptcy, 6 Long Depression	Mid-High
US shale oil	2014peak value	1.5-2x	5-10y	Oil prices drop 75%, ~100 bankrupt	mid
AI 2026 May (Wide Caliber R_T, updated)	Current	1.62x	4-15y Mix	?	Baseline
AI 2026 May (strict caliber R_T,	current	2.28x	same	?	baseline

Case	Period	EPR Approximate Estimate	Asset Life	Terminal Crash Characteristics	Similar to AI Degree
updated)					
British Rail	1846 Peak Value	2.5-3x	50+y	Share price down50-80%	High
US Telecom Optical Fiber (New IP/data)	2000 Peak Value	Approximately 3x	15-20y	$2T Market value evaporates , WorldCom goes bankrupt	High

Warning: All historical EPRs are approximate estimates of the same frame, not precise measurements of "the same caliber". Read several key conclusions from the updated table:

Conclusion 1: The AI position has moved down from the historical bubble peak range to the middle. Early EPR 2.5-3x estimates put AI looking on par with 1846 railroads and 2000 telecoms. Updated 1.6-2.3x puts the AI position between 1873 US Rail (1.2x) and 1846 UK Rail (2.5-3x), and comparable to 2014 US Shale Oil (1.5-2x). This reflects the true impact of 2026 R_T's rapid expansion in the first half of the year.

Conclusion 2: AI still far outperforms normal industry benchmarks. Normal industry 0.65-0.95x, AI 1.6-2.3x. Median EPR 2x = 2.5x normal industry benchmark. Still at the "infrastructure bubble level", just no longer at the peak.

Conclusion 3: The direction has not changed, but the degree has slowed down. AI still shows "capital front + demand extrapolation" characteristics. But R_T cashes out faster than expected - this is the most important "escape" path. If Anthropic + Microsoft AI + OpenAI, etc. 2026-2027 continue to expand at the current growth rate, EPR may fall below 1.5x around 2027 and enter the same level as the peak of shale oil.

Conclusion 4: AI’ s comparability with 1846 Railroad and 2000 Telecommunications decreases, and its comparability with 1873 Rail + Shale Oil increases. 1846 Rail and 2000 Telecom are characterized by terminal demand growth that is severely undercapex paced. AI's current situation is different - R_T is expanding at a rate of 100%+. This reduces the probability of the AI heading towards a 1846/2000-like full-blown crash.

Conclusion 5: But the capex side is still accelerating. Big-5 2026 capex guidance rose to $750B from $620B at the beginning of the year, with Microsoft FY27 hinting at higher. If capex continues to accelerate + R_T growth slows, EPR will rebound. The current 1.6-2.3x is not steady state.

7. Capex/GDP dual-caliber analysis (updated)

Strictly distinguish between two completely different concepts: (1) capex level as a share of GDP; (2) capex contribution to GDP growth. Mixing is a common mistake.

7.1 Capex level to GDP

Case	Period	Capex Level/GDP (annual Diameter)	Description
British Rail	1847Peak value	7-8%	New capex/GDP for the year
US Telecom	2000Peak Value	Approx.1.2%	BEA
US Housing (Residential Investment Investment)	2005-2006	About 6.3%	Historical Highs
AI USA (Big-5 2026)	2026	About 1.7%	$520B AI Part/GDP $30T
AI USA (all industries 2026)	2026	about 2.0%	including neocloud + Oracle + sovereign
Goldman Estimated AI capex/GDP	Current	About 0.8% (narrow bore)	Goldman Sachs Research
AI Global2026	2026	About0.6-0.7%	$700-750B / Global$110T

AI 2026 US capex/GDP is slightly higher than 2000 telecom peak, and well below 1840s rail. The difference between Goldman's estimate (0.8%) and this article's estimate (~1.7-2.0%) lies in the caliber: Goldman uses a more stringent definition of pure AI capex, while this article uses the hyperscaler total capex × 75% AI ratio.

7.2 Capex Contribution to GDP Growth

This is a completely different metric than the capex/GDP level. GDP growth contribution looks at the year-on-year increase in capex, not the total amount. The latest external estimates differ sharply:

Caliber	Estimate	Source
Jason Furman: Information Processing Equipment + Software accounts for H1 2025 GDP Growth	92%	X Posts
St Louis Fed: AI related investment in Q3 2025 real GDP growth	39%	St Louis Fed On the Economy Jan 2026
Goldman Sachs: AI capex contribution to GDP growth (capex incremental caliber)	1.5 pp nominal	Goldman Sachs Research
	0.1-0.2 pp	Jan Hatzius

Caliber	Estimate	Source
Goldman Sachs: AI capex Net GDP growth contribution		

Caliber	Estimate	Source
(After deducting imported capital goods)		
These numbers differ by a factor of nearly 1,000 (0.1-0.2 pp to 92%). Reason:		

- Furman used the NIPA category (Information processing equipment + software). This category H1 2025 has an unusually high contribution to GDP growth due to the surge in AI investment, but the denominator (total GDP growth) is also unusually low. This is the denominator effect.
- St Louis Fed uses more precise AI-related subcategories, resulting in 39% (more credible, but still nominal).
- Goldman further deducts imported capital goods (most imports of GPUs, HBMs, servers) to get net 0.1-0.2 pp. GDP measures domestic production and imported goods do not count towards GDP growth.
- This article adopts Goldman's methodological revisions. The true AI capex's net contribution to U.S. GDP growth may be only 0.1-0.5 pp, far less than the horizontal share (1.7-2.0%) would lead one to think. This is because AI investments 70%+ are imported capital goods (NVIDIA is made with TSMC, HBM is from SK Hynix and Samsung). But this correction in turn means that if AI capex slows down, the direct drag on GDP growth will be smaller (0.1-0.5 pp reversal). Reflexivity is more reflected in stock market valuation → wealth effect → consumption rather than direct GDP calculation.

8. Sensitivity analysis

The EPR center estimate 1.8-2.0x appears to be precise, but actually relies on multiple uncertain parameters. The table below shows the impact of changes in key parameters on EPR.

8.1 Key Parameter Sensitivity (Updated Baseline)

Assumption changes	P change direction	Impact on wide-aperture EPR (baseline1.62x)
R_T raised from $225B to $300B (such as 2027 continues to grow at a high rate)	EPR reduces to 1.21x	
R_T raised from $225B to $400B	unchanged	EPR dropped to 0.91x (close to normal industry)

Assumption changes	P change direction	Impact on wide-aperture EPR (baseline1.62x)
R_T downgraded from $225B to $180B	unchanged	EPR upgraded to 2.02x
R_T Downgraded from $225B to $160B (strict caliber)	Unchanged	EPR Upgraded to 2.28x
K Upgraded from $1.3T to $1.7T (end of 2027)	P Upgraded to approximately $112B	EPR Upgraded to 2.12x
K Upgraded from $1.3T to $2.0T (2028)	P Upgraded to approximately $196B	EPR Upgraded to 2.49x
GPU/network lifetime dropped from 4 years to 3 years	Depreciation increased to approximately $43B	EPR increased to 1.81x
o Upgraded from 10% to 15% (high opex)	opex Upgraded to $65B	EPR Upgraded to 1.91x
Replace r×d with WACC 8%	P up to about $80B	EPR_wacc about 1.97x
EPR 's reasonable 90% confidence interval: 1.0x-2.8x (wide caliber) / 1.5x-3.5x (strict caliber). The point estimate is 1.62x or 2.28x, but the actual values are spread over the interval.		

8.2 EPR Time Path (Updated)

The current K=$1.3T is still below steady state. Estimated by annual capex run rate $600B and L=6.1 years, steady state K=$3.66T. K is still expanding upward. The following is a hypothetical trajectory of the third-speed R_T speed increase (wide aperture $225B starting point). Assumption: annual capex remains at $600B (2026 actual guidance level); other parameters remain unchanged.

Year	K (beginning of the year)	P	R_T 50%CAGR	EPR	R_T 25%CAGR	EPR	R_T 10%CAGR	EPR
2026	$1.3T	$364B	$225B	1.62x	$225B	1.62x	$225B	1.62x
2027	$1.7T	$476B	$338B	1.41x	$281B	1.69x	$248B	1.92x
2028	$2.0T	$560B	$506B	1.11x	$352B	1.59x	$272B	2.06x
2029	$2.3T	$644B	$759B	0.85x	$439B	1.47x	$300B	2.15x
2030	$2.5T	$700B		0.61x	$549B	1.27x	$330B	2.12x

Year	K (beginning of the year)	P	R_T 50%CAGR	EPR	R_T 25%CAGR	EPR	R_T 10%CAGR	EPR
			$1139B					
2031	$2.7T	$756B	$1708B	0.44x	$686B	1.10x	$363B	2.08x
2032	$2.9T	$812B	$2562B	0.32x	$857B	0.95x	$399B	2.03x

Interpretation:

- 50% CAGR (close to 2026 actual first half rate): R_T continues to grow at current pace, EPR fell below 1.5x at 2028, and 2030 is close to normal industry benchmarks. Under this path, the AI cycle achieves "rapid digestion".
- 25% CAGR (still very aggressive): EPR peaked at 2028-2029 around 1.7x, then fell back. 2032 Entering the upper edge of the normal industry. This is a neutral scenario.
- 10% CAGR (conservative): EPR continues at 2x platform. Requires capex to actively slow down to balance. May cause hyperscaler capex downward revision cycle.

Note that 50% CAGR is not extreme. Anthropic's trailing 12-month ARR went from $9B → $40B (400%+). OpenAI 2025 changes from $6B → $20B (230%) throughout the year. If the two companies continue to grow at 50-100% for a few years, it is entirely possible that R_T will reach $500-600B in 2028.

But of course there are risks. Reasoning price wars, open source model challenges, and enterprise penetration saturation may bring R_T's growth to a sudden halt. If R_T growth falls to 10% CAGR and capex continues to be invested at $600B/year level, EPR will continue at the 2x platform in 2027-2029, similar to the digestion period after the establishment of historical bubbles.

9. Key differences between AI and historical bubbles (updated)

9.1 Factors that buffer AI cycles

- Hyperscaler’ s own strong cash flow. Amazon/Microsoft/Google/Meta combined annual free cash flow $300B+.
- Upstream revenue is solid and verifiable. NVIDIA Q1 FY27 DC revenue $75.2B (+92% YoY), forward annualized ~$300B+. Q2 Guidelines $91B/Q.
- R_T Cash out quickly (most important update). Anthropic went from $9B to $40B (6 months), Microsoft AI ARR $37B, OpenAI ~$25B. AI revenue is cashing out faster than any infrastructure cycle in history.

- Diversification reduces systemic risk.
- Real advancement of technology. The cost of inference 2023-2026 has been reduced by 10-30 times cumulatively. Hardware is iterated on an 18-24 month cadence.

9.2 Factors that amplify AI cycles

- OpenAI commitments stack extreme (raised to $1.4T). vs 2026 ARR ~$25B = 56x commitments/ revenue ratio. This is unprecedented in history.
- Revolving financing is huge. NVIDIA 2026 invests $40B+ in equity bets. NVIDIA’ s single $30B investment in OpenAI. NVIDIA + CoreWeave + OpenAI Triangle Loops scale.
- Neocloud local high leverage. CoreWeave includes lease leverage of approximately 4.2x annualized revenue, backlog $99.4B (vs Q1 revenue $2.08B = 48x).
- Rapid depreciation over GPU life. H100's economic value declined significantly within 24 months of B200/Rubin's emergence.
- Open source models challenge proprietary API price tiers. DeepSeek, Llama, Qwen directly threaten proprietary API pricing. This is a factor that 2000 Telecom does not have.
- Microsoft Copilot penetration remains low (3.3-4.4%). Even though AI ARR $37B seems to be high, the coverage rate is still low compared to the Microsoft 365 commercial base (approximately 400 million seats), which means there is a lot of upside space and also means that R_T's expansion depends on the continued increase in penetration rate.

9.3 Comprehensive judgment (updated)

AI cushioning factors are stronger than historical bubbles, and R_T has shown to cash out much faster than 1846 Railroads or 2000 Telecoms. AI amplification factors are also more extreme than historically, especially the size of the OpenAI commitments stack and recurring funding.

Most likely path (updated):

- Anthropic + Microsoft AI maintain rapid growth → R_T continues to expand → EPR further down in 2027-2028
- Dropped to 1.0-1.5x range
- At the same time, capex continues to accelerate (Big-5 capex is still being revised upward) → K is also growing rapidly → EPR will not decline quickly
- OpenAI is the most vulnerable point in the industry chain. Commitments 56x The revenue ratio cannot be absorbed by revenue expansion alone and requires continued equity financing or commitment write-downs.

- Neocloud (CoreWeave, etc.) is also vulnerable. Backlog 48x Quarterly revenue ratio relies on backlog realization. If big clients like Anthropic or Meta renegotiate, it could trigger a chain of events
- Hyperscaler does not have a total collapse. Strong cash flow + diversification.

NVIDIA won't go bankrupt, but the chance of significant compression in forward multiples remains. Its supply commitments $95B and $40B+ investments in customers are so reflexive that any customer-level slowdowns are immediately transmitted to NVIDIA itself.

10. Leading indicators

Monitor high-priority signals for cycle turning points (sorted by importance, adjusted based on updated circumstances):

- 1. Anthropic and OpenAI growth rate. The current 6-week doubling is an anomaly. Any deceleration to monthly growth <10% is a cycle turning signal.
- 1. Hyperscaler asset life policy. Amazon has revised it down from 6 years to 5 years. Microsoft/Google/Meta follow-through will trigger a 10-15% GAAP depreciation upward revision.
- 1. OpenAI commitments fulfillment speed. Whether OpenAI can maintain commitments stack schedule deployment. If OpenAI and Oracle, AWS, CoreWeave, etc. start to delay or renegotiate, it will be the earliest manifestation of the pressure on the commitments stack $1.4T.
- 1. Neocloud refinancing spread and backlog quality. CoreWeave backlog $99.4B looks thick, but is customer focused. Any customer default or delayed deployment would immediately hit EBITDA.
- 1. Hyperscaler capex guidance tone. Changes in rhetoric from "increasing investment" to "improving efficiency".
- 1. Changes in NVIDIA supply commitments. Current $95B. The significant month-on-month decline means that NVIDIA has less confidence in its forward orders.
- 1. AI inference gross profit margin. OpenAI/Anthropic inference gross margin trend. Price war suppressing gross profit improvement is an early warning.
- 1. GPU rental price. H100/H200/B200 spot price. H100 spot Chronically below $3/hour pressure CoreWeave. B200 New cluster breakeven $5/hour.
- 1. AI capex in GDP and contribution to GDP growth. Current capex/GDP is about 1.7-2.0% (US). The ratio continues to rise, which means the economy’ s reliance on AI reflexivity deepens.
- 1. Capital market AI theme valuation quantile. Anthropic current $850-900B valuation (21-23x ARR); OpenAI $852B (34x ARR). The downward movement of these multiples is a periodic signal.

The red combination (simultaneous appearance means that the probability of cyclical turning point increases significantly): Anthropic + OpenAI growth rate deceleration + capex still upward revision + GPU rental price decrease + financing spread expansion + neocloud backlog quality decrease + model company commitments renegotiated.

11. Final judgment

Data available as of 2026-05-25:

Under the assumptions of this article (K=$1.3T, L=6.1 years, R_T=$160-225B), the current EPR_cash of the AI industry chain is in the 1.6x-2.3x range, with a median of about 1.8-2.0x. This is significantly lower than early 2.5-3.0x estimates, mainly driven by 2026 Anthropic ARR surged from $9B to $40B in the first half of the year, Microsoft AI ARR disclosure $37B, and OpenAI maintaining rapid growth. AI current EPR position:

- Significantly higher than normal industry benchmarks (0.65-0.95x) - AI is still in the "infrastructure bubble level" range
- Approximately equal to 2014 US shale oil (1.5-2x) and higher than 1873 US rail aggregate caliber (1.2x)
- Significantly lower than 1846 UK Rail Peak (2.5-3x) and 2000 US Telecom New Section (approximately 3x)
- AI failure modes compared with history:
- Unlike 2007 US housing (price + credit + securitization structure)
- Unlike 1989 Japanese asset bubble (pure price bubble)
- Similarity to 1840s Railroads and 2000 Telecoms decreases as R_T cashes out much faster than those cycles
- Rising similarity to 1873 U.S. railroads and 2014 shale oil

AI capex/GDP ratio:

- US capex level/GDP around 1.7-2.0% (medium-high, but well below 1840s rail 7-8%)
- Net contribution to GDP growth 0.1-0.5 pp (Goldman adjusted for imported capital goods)
- Nominal GDP growth contribution caliber can reach 39% (St Louis Fed)

Most likely path:

- If Anthropic + Microsoft AI + OpenAI, etc. maintain 25%+ annual growth rate, EPR will further decline in 2027-2028, and the AI cycle will achieve "rapid digestion"
- If R_T growth falls below 10% and capex continues at $600B+/year, EPR rebounds above 2x
- Neocloud (CoreWeave, etc.) and over-committed model companies (OpenAI 56x commitments/revenue ratio) remain the most vulnerable
- Hyperscaler has strong cash flow and will not completely collapse

- NVIDIA won't go bankrupt, but massive compression in forward multiples is still possible

One sentence summary: Under the stress test assumptions of this article, AI is a real technology cycle + historical capital front-end + local high leverage + reflexive financing. The order of magnitude of EPR has moved down from 2.5-3x to 1.6-2.3x, which is a true reflection of the faster-than-expected realization of R_T in the first half of 2026. However, AI is still 2-3 times farther away from the normal industry, the capex side is still accelerating, and the OpenAI commitments stack has reached a historically unprecedented level of $1.4T. The most likely outcome is the combination of "partial clearing + soft digestion" in 2027-2029, not a total collapse; but investors holding partial high-leverage links (neocloud, over-committed model companies, long-tail application layers) will bear substantial losses. The probability of a systemic financial crisis is low, but the probability of partial losses of equity and credit is high.

Confidence summary:

- "AI EPR_cash median 1.8-2.0x": Medium (R_T key uncertainty reduced but still present, especially Anthropic
- $40B not audited)
- "AI EPR far exceeds normal industry benchmarks": High
- "AI magnitude is within historical bubble range": High
- "AI orders of magnitude between 1873 railroad and 1846 railroad": Medium-High
- "AI is not like 2007 housing or 1989 Japan": High
- "Hyperscaler will not completely collapse" Will not completely collapse: Medium High
- "OpenAI is the most vulnerable point in the industry chain": high (commitments 56x revenue)
- "Specific clearing time point": low

Twelve, main caveats

- EPR is not a traditional leverage ratio. is the capital service pressure ratio, which is essentially a simplified form of Jorgenson's capital service cost theory.
- EPR is not a complete profit margin indicator. Does not include R&D, SG&A, sales, customer service, equity incentives, taxes. Real profit pressure is higher than EPR suggests.
- EPR > 1 is not the same as "the industry is not profitable". But the terminal revenue is not enough to cover the annual infrastructure pressure defined in this article.
- Historical EPR is an approximation of the same frame. It is not an accurate measurement of "same caliber". All historical EPR confidence low-medium.

- R_T is the most sensitive variable in the full text. Anthropic $40B comes from TechCrunch internal information sources (medium confidence); OpenAI ARR, Microsoft AI ARR, etc. are selectively disclosed by the company and may not be audited.
- K = $1.3T is the median estimate. Cumulative AI capex stock estimates for hyperscaler/neocloud/other entities are subject to ±20% uncertainty.
- Normal industry EPR is an aggregate estimate. The leader in the industry may have a lower EPR and a higher long tail.
- OpenAI commitments $1.4T contain some overlap. Oracle deal and Stargate partially overlap; Anthropic's Amazon $100B purchase commitment is 10 years cumulative; time windows are inconsistent.
- Anthropic ARR $40B is TechCrunch citing internal information. The latest official public figure from the company is $30B (April 2026). "$40B" should be understood as a non-audited private market measure.
- AI capex/GDP figures vary widely from source to source. Goldman 0.8% (narrow aperture) vs. this article's 1.7-2.0% (wide aperture includes all hyperscaler AI related capex).
- The contribution of AI capex to GDP growth varies greatly. From Goldman net 0.1-0.2 pp to Furman 92% (H1 2025 NIPA subcategory). This article adopts Goldman’ s adjusted version of imported capital goods.
- Housing EPR ~1x does not mean housing is not dangerous. Housing is a price bubble and the EPR framework does not apply here. 1873 US railroads only use aggregate EPR ~1.2x. There is no reliable new/old line breakdown data. The mechanism of "reflexivity" itself is not directly quantifiable. NVIDIA invests in OpenAI, and OpenAI buys NVIDIA GPU. The scale of this chain is estimated to be $40B+. The nominal value is a reasonable estimate based on public disclosure, not an audited figure.
- This article provides a research framework and stress test and does not constitute any recommendation to buy or sell securities.

Thirteen, main sources (all updated to 2026-05-25)

AI company financial reports and official disclosures (high confidence):

- NVIDIA Q1 FY27 8-K, CFO commentary, 10-Q (filed 2026-05-20)
- NVIDIA Q4 FY26 CFO commentary (supply commitments $95.2B disclosure)
- Microsoft FY26 Q3 Earnings Call Transcript and 10-Q (2026-04-29, AI ARR $37B)
- Alphabet/Amazon/Meta Q1 2026 10-Q (filed 2026-04-30)

- CoreWeave Q1 2026 Earnings Call Transcript and 10-Q (2026-05-08, backlog $99.4B)
- Anthropic official announcement (2026-04-06, "run-rate revenue has now surpassed $30 billion")
- Anthropic Series G $30B raised at $380B valuation (Feb 2026)
- Anthropic-Google-Broadcom 3.5 GW TPU deal (Apr 2026)
- OpenAI March 2026 $122B funding round at $852B valuation
- AI related industry analysis (medium confidence):
- Goldman Sachs Research: AI capex baseline $765B for 2026 (2026-04)
- Goldman Sachs Research: AI capex GDP impact 0.1-0.2 pp net (Jan Hatzius, 2026-02)
- CreditSights: Hyperscaler Capex 2026 Estimates $750B (2026-02)
- St Louis Fed: AI investment contribution to Q3 2025 GDP growth 39% (Jan 2026)
- Jason Furman: H1 2025 GDP growth contribution analysis
- Bloomberg: AI Circular Deals visualization (2026-03)
- Media coverage (should be downgraded to reported):
- TechCrunch: Anthropic run rate "closer to $40 billion" per internal sources (2026-04-29)
- Reuters: OpenAI/Oracle $300B deal report (2025-09 via WSJ)
- Reuters: Anthropic Google $40B investment (2026-04-24)
- Financial Times: Anthropic potential $40-50B raise at $850-900B valuation (2026-05-07)
- CNBC: NVIDIA $40B+ equity bets in 2026 (2026-05-09)
- VentureBeat: Anthropic $30B ARR with 80x growth (2026-04)

Historical bubble:

- Andrew Odlyzko, "Collective Hallucinations and Inefficient Markets: The British Railway Mania of the 1840s" (2010)
- Andrew Odlyzko, "The Railway Mania of the 1860s"
- Mitchell, B.R., British Historical Statistics
- Williamson, J.G., American Growth and the Balance of Payments 1820-1913
- "Dark Fiber—an Archeology of the Dot-Com Bubble" (Technostatecraft, 2025)
- "Lessons from History: The Rise and Fall of the Telecom Bubble" (Fabricated Knowledge, 2023)

- Global Crossing bankruptcy filings (2002)
- WorldCom Form 10-K 2001 (SEC EDGAR)
- FRED HHMSDODNS, DPI (US housing leverage)
- Federal Reserve History: "The Great Recession and Its Aftermath"

Framework and methodology:

- Jorgenson, D.W. (1963), "Capital Theory and Investment Behavior"
- Public utility revenue requirement framework (standard regulatory economics)
- Damodaran, A. (2024-2025), "Implied required revenue to justify AI capex" blog series